Kirthikaa_Day_02_Networking

THEORY (10 marks)

1. What is a computer network?
   - Computer network is connection of digital telecommunication which allows nodes to share resources
2. What is the difference between a client and a server?
   - A client is a server made available by a Server
   - A Server provides functions or services for clients
3. What is a LAN?
   - LAN – Local Area Network
   - Used for small geographical area network
4. What is a WAN?
   - WAN – Wide Area Network
   - Used for Large Area Network within a certain range but bigger than LAN
5. What is the purpose of a switch?
   - It usually has many interfaces / ports for end hosts usually [more than or equal to 24]
   - Provides connectivity within same LAN
6. What is the purpose of a router?
   - Fewer network interface than switches
   - Provides connectivity between LANs
   - Used to send data over the Internet.
7. What is a firewall?
   - Used for monitoring and controlling network traffic based on configured rules
   - It can be placed either inside/ outside the network as well can be placed on both
   - It is also known as "Next-Generation Firewalls" which is modern also comes with advanced filtering capabilities
8. What is an IP address?
   - IP address is also known as Internet Protocol address
   - Private: Used inside local networks
   - Example: IPv4: 192.168.1.10
9. What is a MAC address?
   - A unique 12-digit physical hardware number assigned by a manufacturer to device network adapter
   - MAC stands for Media Access Control address.
   - Used for communication on Local network segment.
10. What is a default gateway?
    - The node or router IP address device used to send data to outside networks or Internet when destination is not local network
    - Usually at home: 192.168.1.1
11. What is DNS?
    - DNS stands for Domain Name System.
    - It acts as phonebook for internet by encrypting human readable website into machine readable IP address.
12. Why is networking knowledge important for cybersecurity?
    - Without understanding how data normally moves within device it is not possible to secure, monitor or defend a system.
    - It is known as the foundational base for the cybersecurity

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YOUR NETWORK INFORMATION (4 marks)

13. Write your computer's private IPv4 address.
    - IPv4 Address: 10.101.146.110.
14. Write your default gateway
    - Default Gateway: 10.101.128.1
15. Write your DNS server.
    - DNS Servers: 9.9.9.9

SCREENSHOTS (3 marks)

Attach screenshots of

Ipconfig:

```
C:\Windows\System32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1b54:823e:5810:faec%5
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%15
   IPv4 Address. . . . . . . . . . . : 192.168.150.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%2
   IPv4 Address. . . . . . . . . . . : 192.168.35.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:
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```
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::450d:aba6:dee6:4cd7%18
   IPv4 Address. . . . . . . . . . . : 10.101.146.110
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Default Gateway . . . . . . . . . : 10.101.128.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Kirthikaa_Day_02_Networking

Ping

```
C:\Windows\System32>ping 10.101.128.1

Pinging 10.101.128.1 with 32 bytes of data:
Reply from 10.101.128.1: bytes=32 time=17ms TTL=254
Reply from 10.101.128.1: bytes=32 time=10ms TTL=254
Reply from 10.101.128.1: bytes=32 time=3ms TTL=254
Reply from 10.101.128.1: bytes=32 time=5ms TTL=254

Ping statistics for 10.101.128.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 17ms, Average = 8ms

C:\Windows\System32>
```

tracert

```
C:\Windows\System32>tracert 1.1.1.1

Tracing route to one.one.one.one [1.1.1.1]
over a maximum of 30 hops:

  1     9 ms     7 ms     7 ms  10.101.128.1
  2    29 ms     6 ms     2 ms  192.168.200.1
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *     ^C
C:\Windows\System32>
```